

TEM

Training & Evaluation Module

EVA Machine Learning Reasoning Layer — Module Specification

Version 1.0 | References: EVA-ML-Master.docx

1. Purpose

The Training & Evaluation Module trains the candidate learning approaches selected by the MCG and evaluates their performance, stability, and generalization reliability. The module does not aim to maximize performance metrics alone. Its objective is to determine which model can be safely trusted to support decisions.

Core Principle

High accuracy does not mean a useful model. EVA evaluates models behaviorally — a model is considered good because it behaves consistently and responsibly across situations, not because it predicts well once.

2. Three Evaluation Criteria

Every model must be evaluated across three dimensions simultaneously. A model that fails any one of them is rejected.

Criterion	Definition
Performance	How accurately does the model predict on unseen validation data using the chosen metric?
Stability	Does the model produce consistent predictions across different training splits and small input variations?
Generalization	Does the model maintain reasonable performance on the completely held-out test set?

3. Inputs from Global Analysis Ledger

- `model_definition_record` — learning view, split strategy, class imbalance flag
- `candidate_model_record` — modeling approaches, evaluation metric

4. Training Procedure

For each candidate: train on the training set, evaluate on the validation set, perform repeat validation checks across different splits, and record all behavior. Training must be fully reproducible using the same data and configuration stored in the GAL.

5. Performance Evaluation

The module calculates performance using the metric specified in the `candidate_model_record`. Performance evaluation includes validation performance, holdout test performance, and comparison to the interpretable baseline.

A candidate is rejected if its performance is worse than the baseline or performance is inconsistent across validation runs.

⚠ Class Imbalance

If the class imbalance flag is set in the `model_definition_record`, raw accuracy must not be used as the primary metric. The module must use the metric specified by the MCG (F1, AUC-ROC, precision/recall).

6. Overfitting Detection

The module checks whether a model memorizes instead of learning. Indicators include a large gap between training and validation performance, and a sudden performance drop on unseen data. If detected, the model is marked unreliable and rejected.

7. Stability Testing

The module evaluates how sensitive a model is to small changes by testing across different training splits and small input variations. A stable model should produce similar predictions across reasonable variations. Highly unstable models are rejected even if they are accurate.

8. Generalization Assessment

The module evaluates model performance on the holdout test set — data the model has never seen. A model must maintain reasonable performance without drastic degradation. If a model only works on known data, it is rejected.

⚠ Holdout Integrity

The holdout test set must never be used during training or hyperparameter selection. It is used exactly once, for final generalization assessment.

9. Model Comparison & Selection

After evaluation, all candidates are compared across performance, stability, generalization, and interpretability. The highest score alone does not win. The selected model is the most reliable model, not the most complex. A simpler reliable model is preferred over a marginally better complex one.

9.1 When All Candidates Fail

If all candidates are rejected, the module logs the reasons in the `session_log` and halts the pipeline. The MCG and PFTDC are notified to re-evaluate the problem framing or feature set. The pipeline does not retry automatically — a new analysis session is required.

10. Output: model_evaluation_record

Field	Contents
Training Results	Performance metrics per candidate on validation set
Overfitting Assessment	Whether memorization occurred and evidence
Stability Assessment	Sensitivity to variation across splits
Generalization Results	Performance on holdout test set
Selected Model	Chosen candidate and full reasoning
Rejected Models	Why each was rejected, with evidence

11. Operational Rules

1. A model cannot be selected using performance score alone.
2. The holdout test set must not be used during training or tuning.
3. All evaluations must be recorded in the model_evaluation_record.
4. Unstable models must be rejected regardless of their performance score.
5. A simpler reliable model may be preferred over a more complex one.
6. If all candidates fail, halt and log. Do not retry without a new session.
7. Class imbalance flag must be respected in metric interpretation.

12. Relationship to Other Modules

Module	Relationship
MCG	Provides candidate set and evaluation metric
RVEM	Receives selected model for reasoning and leakage validation
PMDD	Deploys the final validated model
Master PRD	Defines global selection principles and failure recovery rules